

Program : Diploma in Chemical Engineering	
Course Code : 4072	Course Title: Organic Chemical Technology
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To interpret the industrial processes for the manufacturing of organic chemicals.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic Chemistry		Applied chemistry	1
		Applied chemistry II	2
Unit operations and Unit process		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the manufacturing of Soaps and Oils.	12	Understanding
CO2	Summarize the manufacturing of Sugar, paper and Starch.	18	Understanding
CO3	Summarize the manufacturing of leather, fibers, rubbers and pharmaceuticals	18	Understanding
CO4	Summarize the manufacturing of Explosives, Propellants, Perfumes and Adhesives.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the manufacturing of Soaps and Oils.		
M1.01	Outline the importance of organic chemicals.	2	Understanding.
M1.02	Summarize oils and fats.	5	Understanding
M1.03	Summarize the manufacturing of Soaps.	3	Understanding.
M1.04	Summarize the manufacturing of detergents.	2	Understanding.
Contents: Introduction to organic technology - Significance of organic chemicals – List the major organic chemical manufacturing industries in Kerala. Oils - Edible and non edible oils – Fats - Fatty acids - Oils – List common oil seeds - Extraction of oils - Expeller pressing - Solvent extraction using hexane with flow diagram – Processing of oil - Refining – Bleaching – Hydrogenation - Deodorization - Chemistry of saturated and unsaturated fats and oils – Properties of oil - Acid value - Saponification value - Iodine value - Rancidity of an oil. Soaps – Classification hard soap and soft soap – Manufacturing of soap by hot process (full boiling) with flow diagram - Raw materials – Saponification - Glycerin recovery - Drying and finishing - Additives of soap - Glycerin purification. Detergents – Types of detergents - Chemistry of detergents - Alkyl benzene sulphonate – Manufacturing of Alkyl benzene sulphonate with flow diagram - Biodegradability of detergents, Comparison of detergent with soap - Manufacturing process of detergent through spray drying process with raw materials and flow diagram.			

CO2	Summarize the manufacturing of Sugar, paper and Starch.		
M2.01	Illustrate the manufacturing of sugar.	4	Understanding
M2.02	Illustrate the manufacturing of alcohol.	4	Understanding
M2.03	Illustrate the manufacture of pulp and paper.	4	Understanding.
M2.04	Illustrate the manufacturing of starch.	5	Understanding
	Series Test – I	1	

Contents:

Sugar manufacturing - Raw material - Chemical reaction - Manufacturing process including flow diagram - Raw material preparation - Concentration - Crystallization - Refining of sugar – Byproducts - Bagasse and molasses.

Ethyl alcohol - Manufacturing of ethyl alcohol from molasses - Manufacturing process including flow diagram - Raw material preparation – Fermentation - Distillation and rectification.

Pulp and paper - Raw materials – Types of pulping, chemical, mechanical and semi chemical methods – Manufacturing of wood pulp by Sulphate process (Kraft) with flow diagram - Bleaching of pulp - Chemical recovery from black liquor - Paper manufacture with process flow diagram – additives and their purpose.

Starch - Manufacture of starch from corn including flow diagram - Raw material preparation – Steeping – Disintegrating – Bleaching - Filtration - Drying - Starch derivatives – Uses of starch derivatives.

CO3	Summarize the manufacturing of leather, fibers, rubbers and pharmaceuticals		
M3.01	Illustrate the manufacturing of leather.	5	Understanding
M3.02	Illustrate the manufacturing of fibers.	4	Understanding
M3.03	Illustrate the manufacturing of rubber.	4	Understanding
M3.04	Illustrate the manufacturing of Pharmaceuticals.	5	Understanding

Contents:

Leather – Preparation of raw hides and skins - Tanning process - Types - Vegetable tanning, Chrome tanning- flow diagram - Finishing operations for leather - Applications.

Fibers – Classification - Cellulosic and Synthetic fibers – Illustrate manufacturing process of viscose rayon and nylon 6, 6.

Rubbers –Natural rubber - Synthetic rubber- Processing of latex – Crepe rubber - Smoked rubber -Vulcanization of rubber - Advantages of vulcanization - Synthetic rubber - Nitrile rubber-Properties and uses - Neoprene rubber - Properties and uses - Butyl rubber – Properties and uses.

Pharmaceutical – Outline of Pharmaceutical industries - Significance – Antibiotics - Manufacturing of penicillin - Flow diagram - Uses.

CO4	Summarize the manufacturing of Explosives, Propellants, Perfumes and Adhesives.		
M4.01	Illustrate the manufacturing of explosives	3	Understanding
M4.02	Illustrate the manufacturing of rocket propellants	3	Understanding
M4.03	Illustrate the manufacturing of perfumes.	3	Understanding
M4.04	Illustrate the manufacturing of adhesives.	2	Understanding
	Series Test – II	1	
Contents: Explosives - Introduction - Classification of explosives - Primary-low explosives - High explosives - List important explosives - Applications. Propellants - General features of propellants - Choice of propellant system - Solid propellants - Nitro glycerin - Liquid propellants – Compare solid propellant versus Liquid propellant. Perfumes, - Perfumery chemicals - Perfumery processes - Production methods of plant oils -Distillation – Expression - Extraction with solvents. Adhesives – Natural and Synthetic adhesives – List adhesives from natural sources – List synthetic adhesives – Illustrate the manufacturing of adhesive from starch – Illustrate the manufacturing of urea formaldehyde.			

Text / Reference

T/R	Book Title/Author
T1	Dryden's Outlines of Chemical Technology; Edited and revised by M. Gopala Rao and Marshall Sittig
R1	Shreve's Chemical Process Industries; G.T. Austin
R2	Chemical Process Technology; O.P. Gupta
R3	Encyclopedia of Chemical Technology ; Kirk-Othmer
R4	A Textbook of Engineering Chemistry; Dr. SS. Dhara and Dr. Umare S chand publishers.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/107/103107082/